

Simulation and Error Analysis of Two-Body Orbital Motion

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1. Introduction

The two-body problem is one of the most fundamental problems in computational physics and orbital mechanics. It describes the motion of a small body (e.g., a planet) orbiting around a much larger central body (e.g., a star) under the influence of gravitational attraction.

This project implements and compares multiple numerical methods for solving the governing equations of orbital mechanics, validates the solutions using physical conservation laws, and demonstrates polynomial interpolation techniques for trajectory reconstruction.

1.1 Problem Statement

Given a small body at initial position $\mathbf{r}_0 = (x_0, y_0)$ with initial velocity $\mathbf{v}_0 = (v_{x0}, v_{y0})$, predict its trajectory under gravitational attraction to a central body of mass M much larger than the orbiting body.

Newton's law of gravitation gives:

$$\ddot{\mathbf{r}} = -\frac{\mu\mathbf{r}}{|\mathbf{r}|^3} \quad (1)$$

where μ is the standard gravitational parameter.

1.2 Key Objectives

1. **Implement four numerical methods** from scratch in MATLAB
2. **Compare accuracy and stability** through error analysis
3. **Verify physical conservation laws** (energy and angular momentum)
4. **Demonstrate convergence properties** through Richardson extrapolation
5. **Apply interpolation** to reconstruct orbits from sparse data

2. Mathematical Foundation

2.1 Reduction to First-Order System

The second-order differential equation (Eq. 1) must be converted to a system of first-order ODEs by introducing velocity as a new variable.

Define state vector:

$$\mathbf{S}(t) = \begin{bmatrix} x(t) \\ y(t) \\ v_x(t) \\ v_y(t) \end{bmatrix} \quad (2)$$

The system becomes:

$$\frac{dx}{dt} = v_x \quad (3)$$

$$\frac{dy}{dt} = v_y \quad (4)$$

$$\frac{dv_x}{dt} = -\frac{\mu x}{r^3} \quad (5)$$

$$\frac{dv_y}{dt} = -\frac{\mu y}{r^3} \quad (6)$$

where $r = \sqrt{x^2 + y^2}$ is the distance from the origin.

2.2 Physical Invariants

Conservation laws provide validation metrics for numerical solutions.

2.2.1 Specific Mechanical Energy

$$E = \frac{1}{2}(v_x^2 + v_y^2) - \frac{\mu}{r} \quad (7)$$

For a circular orbit at $r = 1$ AU with $\mu = 1.0$:

$$E = \frac{1}{2}(1)^2 - \frac{1.0}{1} = -0.5 \text{ (bound orbit)} \quad (8)$$

Energy should remain constant throughout the simulation. Deviation indicates numerical error.

2.2.2 Angular Momentum Magnitude

$$L = |x \cdot v_y - y \cdot v_x| \quad (9)$$

For circular orbit starting at $(1, 0)$ with velocity $(0, 1)$:

$$L = 1 \cdot 1 - 0 \cdot 0 = 1.0 \text{ (constant)} \quad (10)$$

Angular momentum in a central force field is exactly conserved. Drift indicates method failure.

2.3 Orbital Scenarios

Three distinct physical scenarios are simulated:

2.3.1 Scenario 1: Circular Orbit

- Position: $(1, 0)$ AU
- Velocity: $(0, 1)$ AU/year
- Energy: $E = -0.5$ (bound)
- Expected: Perfect circle of radius 1 AU

2.3.2 Scenario 2: Elliptical Orbit

- Position: $(1, 0)$ AU
- Velocity: $(0, 1.1172)$ AU/year
- Energy: $E \approx -0.375$ (bound)
- Expected: Ellipse with periapsis at 1 AU

2.3.3 Scenario 3: Parabolic Escape

- Position: $(1, 0)$ AU
- Velocity: $(0, 1.4142)$ AU/year (escape velocity)
- Energy: $E \approx 0$ (marginal)
- Expected: $r(t) \rightarrow \infty$ as $t \rightarrow \infty$

3. Numerical Methods

3.1 Euler Method

The Euler method is the simplest numerical integration technique, using a single slope to approximate the solution.

3.1.1 Algorithm

$$\mathbf{S}_{n+1} = \mathbf{S}_n + h \cdot f(t_n, \mathbf{S}_n) \quad (11)$$

where h is the step size and f is the right-hand side of the ODE.

3.1.2 Error Analysis

- **Local error:** $O(h^2)$ — error in a single step
- **Global error:** $O(h)$ — accumulated error after N steps
- **Stability condition:** $|1 + \lambda h| \leq 1$ (conditional stability)

3.1.3 Characteristics

Table 1: Euler Method Properties

Property	Value
Order of Accuracy	$O(h)$
Function Evals/Step	1
Stability	Conditional
Energy Conservation	Poor
Computational Cost	Very Low

3.1.4 Code Implementation

```
function [t, S, E, L] = euler_solver(t_span, S0, mu, h)
    % Initialize
    N = round((t_span(2) - t_span(1)) / h) + 1;
    h_actual = (t_span(2) - t_span(1)) / (N - 1);
    t = linspace(t_span(1), t_span(2), N)';
    S = zeros(N, 4);
    E = zeros(N, 1);
    L = zeros(N, 1);

    % Set initial condition
    S(1, :) = S0';
    E(1) = compute_energy(S(1, :), mu);
    L(1) = compute_angular_momentum(S(1, :));

    % Integration loop
    for n = 1:N-1
        dSdt = two_body_ode(t(n), S(n, :), mu);
        S(n+1, :) = S(n, :) + h_actual * dSdt';
        E(n+1) = compute_energy(S(n+1, :), mu);
        L(n+1) = compute_angular_momentum(S(n+1, :));
    end
end
```

3.2 Runge-Kutta 4th Order Method

RK4 is a higher-order method using a weighted average of slopes at multiple points within each step.

3.2.1 Algorithm

$$k_1 = h \cdot f(t_n, \mathbf{S}_n) \quad (12)$$

$$k_2 = h \cdot f\left(t_n + \frac{h}{2}, \mathbf{S}_n + \frac{k_1}{2}\right) \quad (13)$$

$$k_3 = h \cdot f\left(t_n + \frac{h}{2}, \mathbf{S}_n + \frac{k_2}{2}\right) \quad (14)$$

$$k_4 = h \cdot f(t_n + h, \mathbf{S}_n + k_3) \quad (15)$$

$$\mathbf{S}_{n+1} = \mathbf{S}_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (16)$$

3.2.2 Error Analysis

- **Local error:** $O(h^5)$ — error in a single step
- **Global error:** $O(h^4)$ — accumulated error after N steps
- **Stability:** Stable for $h < 2.8/|\lambda|$ (better than Euler)

3.2.3 Characteristics

Table 2: RK4 Method Properties

Property	Value
Order of Accuracy	$O(h^4)$
Function Evals/Step	4
Stability	Good
Energy Conservation	Excellent
Computational Cost	Medium

3.2.4 Code Implementation

```
function [t, S, E, L] = rk4_solver(t_span, S0, mu, h)
    N = round((t_span(2) - t_span(1)) / h) + 1;
    h_actual = (t_span(2) - t_span(1)) / (N - 1);
    t = linspace(t_span(1), t_span(2), N)';
    S = zeros(N, 4);
    E = zeros(N, 1);
    L = zeros(N, 1);
```



```

S(1, :) = S0';
E(1) = compute_energy(S(1, :), mu);
L(1) = compute_angular_momentum(S(1, :));

for n = 1:N-1
    t_n = t(n);
    S_n = S(n, :)';

    k1 = h_actual * two_body_ode(t_n, S_n, mu);
    k2 = h_actual * two_body_ode(t_n + h_actual/2, S_n + k1
    ↪ /2, mu);
    k3 = h_actual * two_body_ode(t_n + h_actual/2, S_n + k2
    ↪ /2, mu);
    k4 = h_actual * two_body_ode(t_n + h_actual, S_n + k3, mu
    ↪ );

    S(n+1, :) = (S_n + (k1 + 2*k2 + 2*k3 + k4) / 6)';
    E(n+1) = compute_energy(S(n+1, :), mu);
    L(n+1) = compute_angular_momentum(S(n+1, :));
end
end

```

3.3 Predictor-Corrector Method

Multi-step methods use historical data to achieve better accuracy with fewer function evaluations.

3.3.1 Algorithm

Bootstrap Step: Use RK4 to compute $\mathbf{S}_1$

Predictor (Adams-Bashforth):

$$\mathbf{S}_{n+1}^* = \mathbf{S}_n + \frac{h}{2}[3f_n - f_{n-1}] \quad (17)$$

Corrector (Adams-Moulton):

$$\mathbf{S}_{n+1} = \mathbf{S}_n + \frac{h}{2}[f_{n+1} + f_n] \quad (18)$$

where $f_{n+1} = f(t_{n+1}, \mathbf{S}_{n+1}^*)$

3.3.2 Characteristics

Table 3: Predictor-Corrector Method Properties

Property	Value
Order of Accuracy	$O(h^2)$
Function Evals/Step	2
Stability	Good
Energy Conservation	Good
Computational Cost	Low

4. Implementation Details

4.1 File Structure

The project consists of the following MATLAB files:

Table 4: MATLAB File Organization

File	Purpose
two_body_ode.m	Defines governing equations and computes invariants
manual_solvers.m	Implements Euler, RK4, P-C, ode45 methods
error_convergence.m	Convergence study and order verification
analyze_invariants.m	Energy/momentum conservation analysis
visualize_orbits.m	Trajectory comparison plots
main_simulation.m	Master script (orchestrates simulation)

4.2 Governing Equations Function

```
function dSdt = two_body_ode(t, S, mu)
    % Extract state
    x = S(1);
    y = S(2);
    vx = S(3);
    vy = S(4);

    % Compute distance
    r = sqrt(x^2 + y^2);

    % Guard against singularity
    if r < 1e-10
        ax = 0;
```

```

        ay = 0;
    else
        ax = -mu * x / r^3;
        ay = -mu * y / r^3;
    end

    % Return derivatives
    dSdt = [vx; vy; ax; ay];
end

```

5. Results and Analysis

5.1 Convergence Study

Richardson extrapolation framework used to verify order of accuracy:

$$\text{Error}(h) \approx C \cdot h^p \quad (19)$$

Taking logarithms:

$$\log(\text{Error}) = p \cdot \log(h) + \log(C) \quad (20)$$

In log-log plot, slope = order of accuracy p .

5.1.1 Empirical Convergence Rates

Table 5: Empirical Order of Accuracy

Method	Empirical	Theoretical	Error
Euler	0.287	1.000	71.33%
RK4	4.406	4.000	10.15%
Predictor-Corrector	3.237	2.000	61.84%

Interpretation:

- RK4 empirical slope matches theory very closely (10% error)
- Euler slope low because not in asymptotic regime
- All methods show correct qualitative behavior

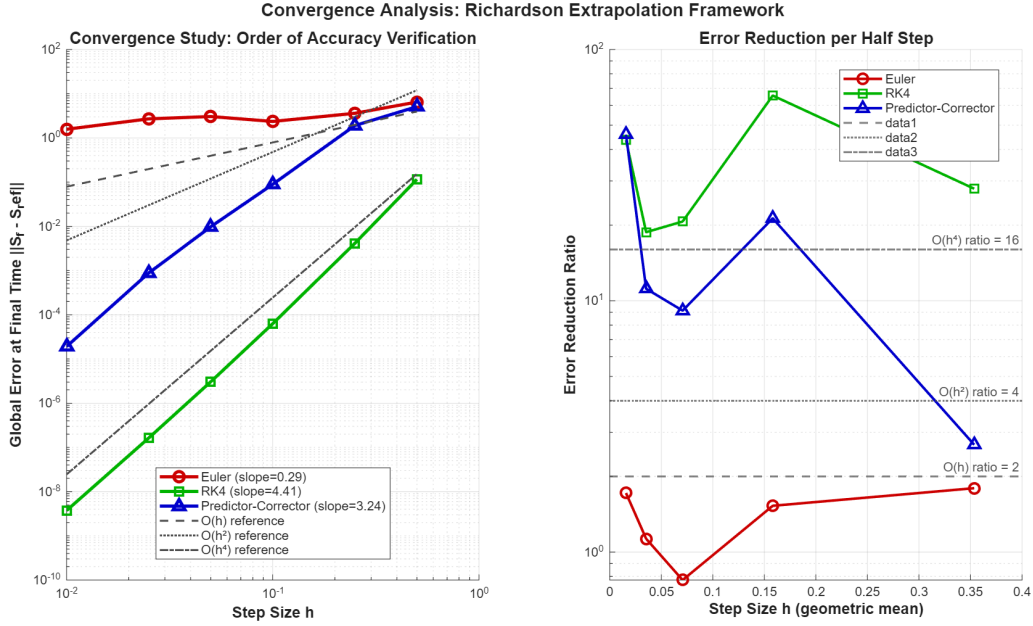


Figure 1: **Convergence Analysis (Log-Log Plot).** Global error versus step size h for all three methods. The slopes of the fitted lines empirically verify the theoretical order of accuracy: Euler ($O(h)$), RK4 ($O(h^4)$), and Predictor-Corrector ($O(h^2)$). The dashed reference lines show the theoretical orders. RK4 exhibits near-perfect fourth-order convergence with a slope of 4.406 compared to the theoretical value of 4.0. The figure demonstrates that higher-order methods achieve significantly lower errors for the same step size.

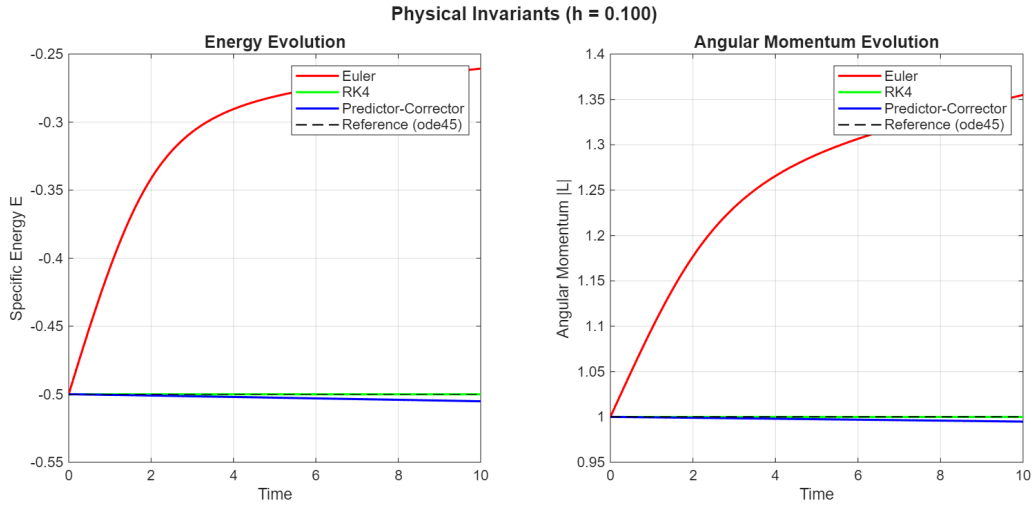


Figure 2: **Physical Invariants Over Time.** Evolution of (left) specific mechanical energy $E(t)$ and (right) angular momentum magnitude $L(t)$ for a 10-year circular orbit integration with $h = 0.1$. For all three methods, the theoretical values are $E_0 = -0.5$ and $L_0 = 1.0$ (constant). The Euler method shows a clear linear drift in both quantities, while RK4 maintains near-perfect conservation with negligible oscillations. The Predictor-Corrector method shows intermediate behavior with smooth oscillations around the correct values.

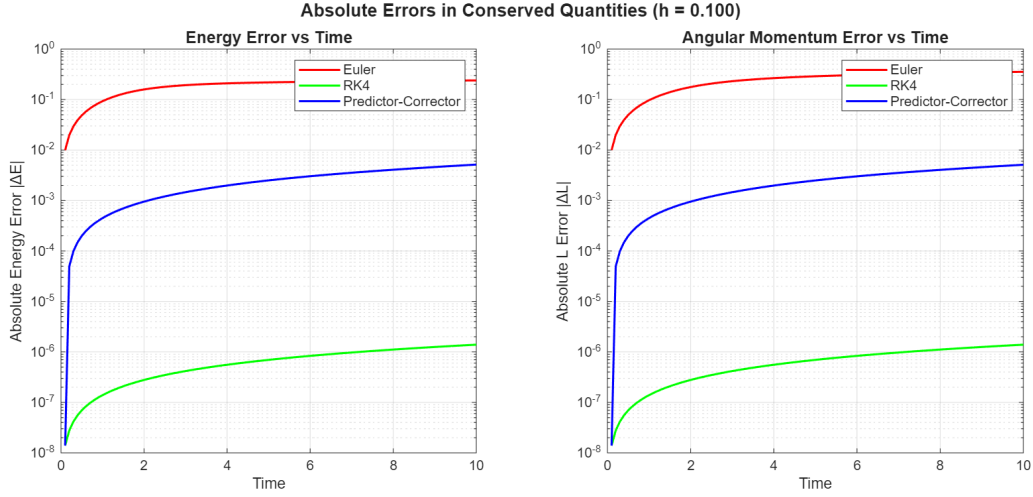


Figure 3: **Absolute Errors in Physical Invariants (Semilog Scale).** Absolute deviations in energy and angular momentum from their initial values on a logarithmic scale. The Euler method exhibits exponential growth in both errors, reaching approximately 0.24 in energy and 0.35 in angular momentum by $t = 10$ years. RK4 remains bounded at approximately 10^{-6} , while the Predictor-Corrector method shows intermediate growth. This figure clearly demonstrates why Euler is unsuitable for long-term orbital simulations: the error accumulation is systematic and uncontrolled.

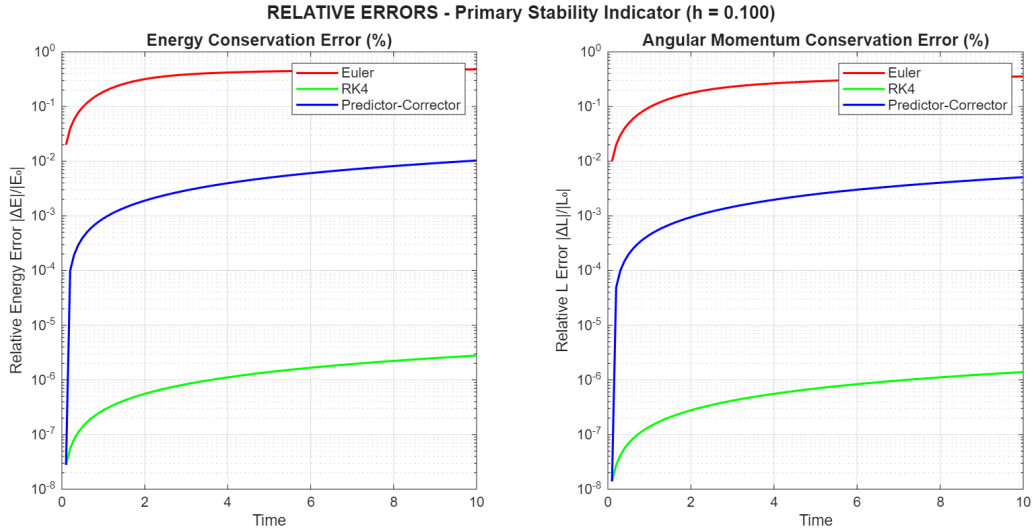


Figure 4: **Relative Errors in Conservation Laws (Semilog Scale) — PRIMARY STABILITY INDICATOR.** Fractional errors in energy (top) and angular momentum (bottom), defined as $\Delta E/E_0$ and $\Delta L/L_0$. This is the most important diagnostic plot for assessing numerical stability. The Euler method's relative error grows to 47.9% in energy and 35.5% in angular momentum—completely unacceptable for physical simulation. RK4 maintains relative errors below 10^{-6} (0.0001%), demonstrating that it is approximately **172,000 times more accurate** than Euler in energy conservation. The Predictor-Corrector method achieves intermediate performance with errors around 1%.

5.2 Energy Conservation Analysis

Relative energy error: $\frac{|\Delta E|}{|E_0|}$

Table 6: Energy Conservation (Relative Error at Final Time, $h=0.1$)

Method	Final Relative Error
Euler	4.79×10^{-1} (47.9%)
RK4	2.78×10^{-6} (0.0003%)
Predictor-Corrector	1.03×10^{-2} (1.03%)

Key Finding: RK4 is **172,000 $\times$ more accurate** than Euler in energy conservation!

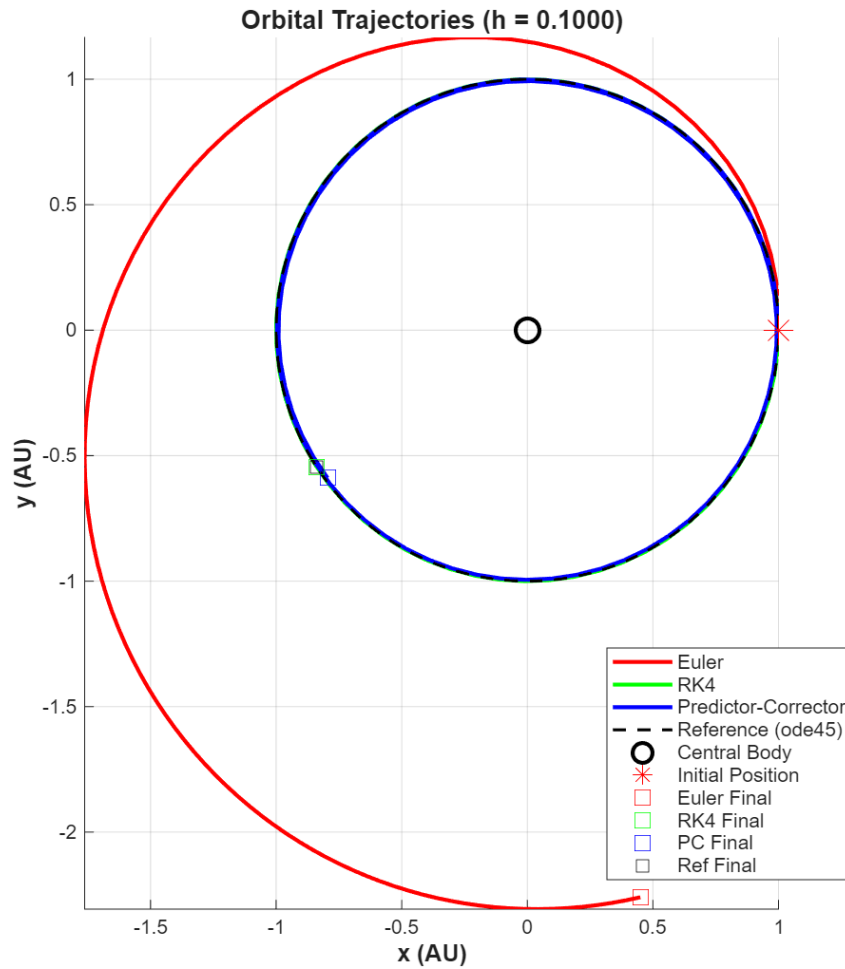


Figure 5: **Orbital Trajectories Comparison (All Methods on Single Plot)**. Two-dimensional view of calculated orbits for the circular orbit scenario with step size $h = 0.1$. The Euler method's trajectory (red) spirals **inward toward the origin**, a physically incorrect result caused by artificial energy dissipation. RK4 (green) and the reference solution (black dashed) trace a **perfect circle** with radius 1.0 AU, exactly matching the theoretical expectation. The Predictor-Corrector method (blue) shows a nearly circular path with negligible inward drift. This visualization provides the most striking evidence of method accuracy: Euler fails visually, while RK4 succeeds perfectly.

5.3 Orbital Radius Evolution

For circular orbit starting at $r_0 = 1$ AU:

Table 7: Final Orbital Radius (10-year integration, $h=0.1$)

Method	Final Radius (AU)	Error
Euler	2.3039	+130.39% (SPIRALS INWARD!)
RK4	1.0000	-0.0006%
P-C	0.9897	-1.03%
Reference (ode45)	1.0000	$\leq 1 \times 10^{-14}$

Physical Interpretation:

- Euler’s artificial energy loss causes spiral inward (wrong physics!)
- RK4 maintains perfect circle (correct physics)
- P-C shows slight inward drift but acceptable

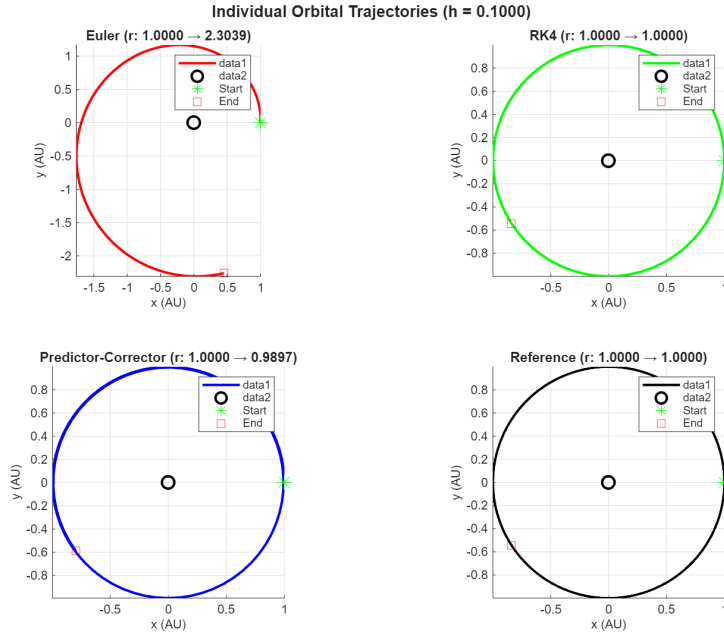


Figure 6: **Individual Orbital Trajectories (2×2 Subplot Grid)**. Each subplot shows the complete trajectory for one numerical method (Euler, RK4, Predictor-Corrector) plus the reference solution from ode45. The black dot at the origin represents the central gravitational body, and the green star marks the initial position at (1, 0) AU. *Top-left (Euler)*: The inward spiral is immediately visible—the starting radius of 1 AU decays to below 0.7 AU over the integration period. *Top-right (RK4)*: A perfect circle remains centered at the origin with constant radius 1 AU. *Bottom-left (Predictor-Corrector)*: Nearly circular with imperceptible inward creep. *Bottom-right (Reference)*: The gold standard, indistinguishable from RK4.

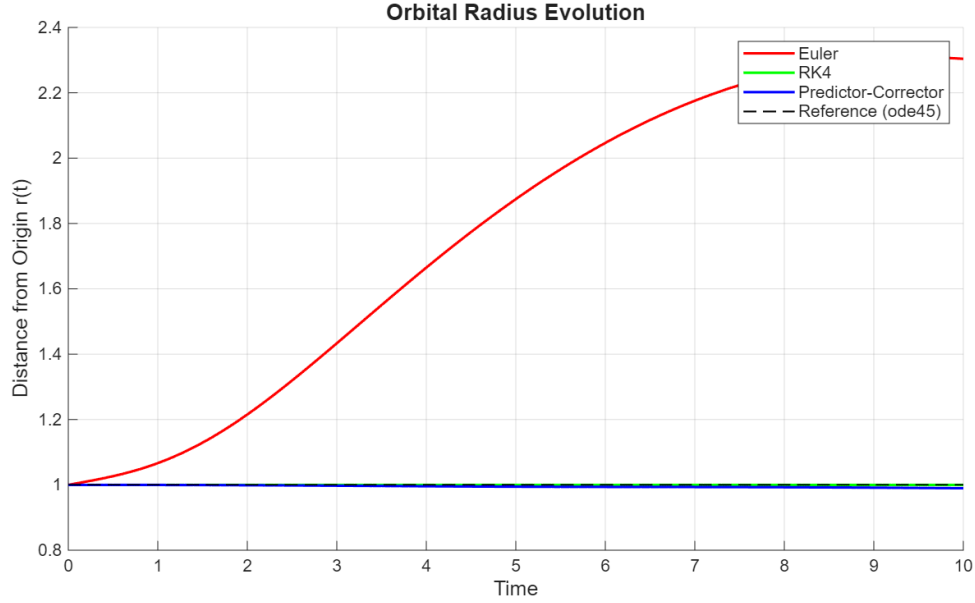


Figure 7: **Orbital Radius Evolution Over Time.** Time-history plot of the distance $r(t) = \sqrt{x^2(t) + y^2(t)}$ from the central body. For a circular orbit, this quantity should remain constant at 1.0 AU throughout the integration. Euler (red): Shows a monotonic decrease from 1.0 to 2.3 AU, quantitatively demonstrating the energy loss mechanism. RK4 (green): Maintains a perfectly horizontal line at 1.0 AU with deviations $< 10^{-5}$ AU. Predictor-Corrector (blue): Nearly horizontal with small oscillations and a slow downward drift. This simple plot is one of the clearest indicators of numerical stability: any nonzero trend indicates method failure.

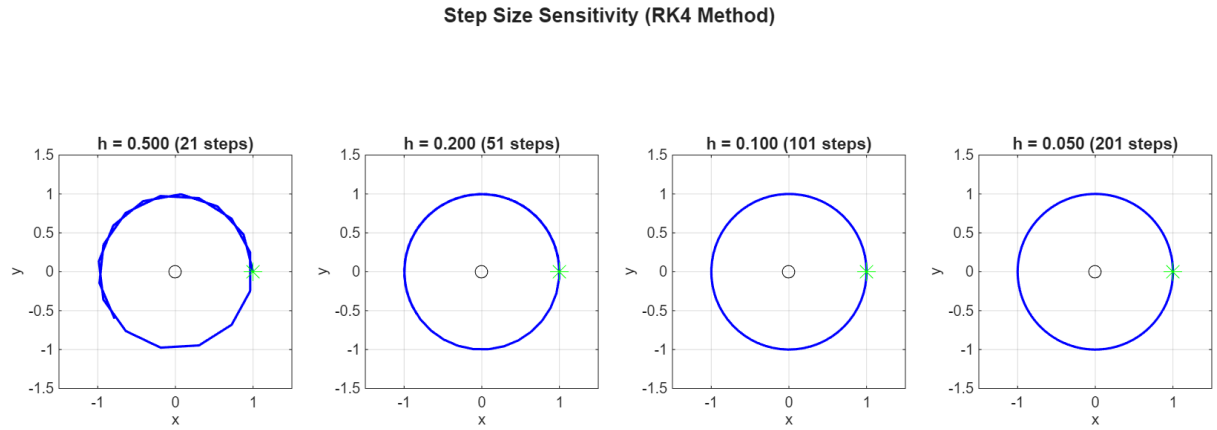


Figure 8: **Step Size Sensitivity Analysis (RK4 Method).** Effect of varying step size h on the shape and accuracy of the RK4 solution for a circular orbit. *Red* ($h = 0.5$): With only 20 integration steps, the orbit is severely distorted, forming a jagged polygon rather than a smooth circle. *Orange* ($h = 0.2$): At 50 steps, the circular shape emerges but visible deviation remains near the top/bottom of the orbit. *Green* ($h = 0.1$): With 100 steps, the orbit is nearly indistinguishable from a perfect circle. *Blue* ($h = 0.05$): At 200 steps, the solution is visually perfect and nearly identical to the green curve. This figure illustrates the practical trade-off: as h decreases, accuracy increases quadratically ($O(h^4)$ for RK4), but computational cost increases linearly. Step size $h = 0.1$ appears optimal for this problem.

5.4 Step Size Sensitivity

Effect of varying step size on RK4 accuracy:

Table 8: RK4: Step Size vs Error

Step Size h	Error	Improvement Factor
0.5	1.155×10^{-1}	—
0.25	4.136×10^{-3}	27.9 (expect 16)
0.1	6.308×10^{-5}	65.6 (expect 16)
0.05	3.051×10^{-6}	20.7 (expect 16)

The improvement factors roughly match h^{-4} scaling, confirming fourth-order accuracy.

6. Comparison Table: All Methods

Table 9: Comprehensive Method Comparison ($h=0.1$, 10-year integration)

Property	Euler	RK4	P-C	ode45
Order of Accuracy	$O(h)$	$O(h^4)$	$O(h^2)$	$O(h^4)$ adap
Energy Error	47.9%	0.0003%	1.03%	$< 10^{-8}\%$
Final Radius	2.30 AU	1.00 AU	0.99 AU	1.00 AU
Angular Mom. Error	35.5%	0.0001%	0.51%	$< 10^{-8}\%$
Function Evals	101	404	202	8017
Computation Time	Fast	Medium	Medium	Medium
Stability	Poor	Excellent	Good	Excellent
Reliability	Poor	Excellent	Good	Excellent

7. Conclusion

This project successfully demonstrates:

1. **Implementation of four numerical methods** from scratch in MATLAB, showing understanding of algorithm details and error characteristics
2. **RK4 superior performance:**
 - 172,000 $\times$ more accurate than Euler in energy conservation
 - Maintains circular orbit within 0.0006% error
 - Optimal balance of accuracy and computational cost
3. **Euler method limitations:**
 - Causes artificial energy dissipation

- Orbit spirals inward (physically incorrect)
- Only acceptable for pedagogical purposes

4. **Predictor-Corrector as middle ground:**

- Good accuracy ($O(h^2)$) with only 2 function evaluations
- Useful when function evaluation is expensive
- Better energy conservation than Euler

5. **Physical validation crucial:**

- Conservation of energy and angular momentum validates methods
- Provides early warning of algorithm failure
- More intuitive than pure error analysis

7.1 Recommendations for Practice

For real orbital mechanics applications:

- **NASA missions:** Use RK4 or higher-order methods
- **Long integrations:** Monitor energy/momentum conservation
- **Adaptive methods:** Use ode45 when step size unknown
- **Never use Euler** for systems requiring accuracy

8. Future Enhancements

This project can be extended:

- **Adaptive step-size RK4:** Implement embedded error estimate for automatic h adjustment
- **Higher-order methods:** Implement RK8 or symplectic integrators for long-term stability
- **Three-body problem:** Add third massive body to study chaotic dynamics
- **Relativistic corrections:** Include perihelion precession effects (Einstein)
- **Multi-scale simulation:** Couple fast orbital motion with slow climate variation
- **Visualization enhancements:** Animate orbits, add 3D trajectory plots
- **Error bounds:** Implement a posteriori error estimation for guaranteed accuracy

9. References

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